

Danh mục: medication

Type 2 diabetes in adults: management

Type 2 diabetes in adults: management

NICE guideline

Reference number:

NG28

Published:

02 December 2015

Last updated:

18 February 2026

Download guidance (PDF)

Blood glucose management

1.5 HbA1c measurement and targets

Measurement

1.5.1

Measure HbA1c levels in adults with type 2 diabetes every:

3 to 6 months (tailored to individual needs) until HbA1c is stable on unchanging therapy

6 months once the HbA1c level and blood glucose lowering therapy are stable.

[2015]

1.5.2

Measure HbA1c using methods calibrated according to International Federation of Clinical Chemistry (IFCC) standardisation.

[2015]

1.5.3

If HbA1c monitoring is invalid because of disturbed erythrocyte turnover or abnormal haemoglobin type, estimate trends in blood glucose control using one of the following:

quality-controlled plasma glucose profiles

total glycated haemoglobin estimation (if abnormal haemoglobins)

fructosamine estimation.

[2015]

1.5.4

Investigate unexplained discrepancies between HbA1c and other glucose measurements. Seek advice from a team with specialist expertise in diabetes or clinical biochemistry.

[2015]

Targets

NICE has produced a

patient decision aid on agreeing HbA1c targets

, which also covers factors to weigh up when discussing HbA1c targets with patients.

1.5.5

Discuss and agree an individual HbA1c target with adults with type 2 diabetes (see recommendations 1.5.6 to 1.5.10).

Encourage them to reach their target and maintain it, unless any resulting adverse effects (including hypoglycaemia), or

Danh mục: medication

their efforts to achieve their target impair their quality of life. Think about using the NICE patient decision aid on weighing up HbA1c targets to support these discussions.

[2015, amended 2022]

1.5.6

Offer advice on healthy living, and medicines, to support adults with type 2 diabetes to reach and maintain their HbA1c target (see the

sections on dietary advice

bariatric surgery

, and

person-centred medicines

, and the

NHS Better Health website

). For more information about supporting adherence, see

NICE's guideline on medicines adherence

[2015, amended 2026]

1.5.7

For adults whose type 2 diabetes is managed either by healthy living and diet, or healthy living and diet combined with an initial medication regimen that is not associated with hypoglycaemia (see the

section on initial medicines

), support them to aim for an HbA1c level of 48 mmol/mol (6.5%). For adults on a medicine associated with hypoglycaemia, support them to aim for an HbA1c level of 53 mmol/mol (7.0%).

[2015, amended 2026]

1.5.8

In adults with type 2 diabetes, if HbA1c levels are not adequately controlled by the initial medication regimen and rise to 58 mmol/mol (7.5%) or higher:

reinforce advice about diet, healthy living and adherence to medicines and

support the person to aim for an HbA1c level of 53 mmol/mol (7.0%) and

intensify medicines.

[2015, amended 2026]

1.5.9

Consider relaxing the target HbA1c level (see recommendations 1.5.7 and 1.5.8 and

NICE's patient decision aid on type 2 diabetes: agreeing someone's blood glucose (HbA1c) target

) on a case-by-case basis and in discussion with adults with type 2 diabetes, with particular consideration for people who are older or frailer, if:

they are unlikely to achieve longer-term risk-reduction benefits, for example, people with a reduced life expectancy

tight blood glucose control would put them at high risk if they developed hypoglycaemia, for example, if they are at risk

of falling, they have impaired awareness of hypoglycaemia, or they drive or operate machinery as part of their job

intensive management would not be appropriate, for example if they have significant comorbidities.

[2015, amended 2022]

1.5.10

If adults with type 2 diabetes reach an HbA1c level that is lower than their target and they are not experiencing hypoglycaemia, encourage them to maintain it. Be aware that there are other possible reasons for a low HbA1c level, for

Danh mục: medication

example deteriorating renal function or sudden weight loss.

[2015]

1.5.11

For guidance on HbA1c targets in pregnancy, see

NICE's guideline on diabetes in pregnancy

[2015]

1.6 Self-monitoring of capillary blood glucose

These recommendations relate to self-monitoring by capillary blood glucose monitoring.

1.6.1

Take the

Driver and Vehicle Licensing Agency (DVLA)'s Assessing fitness to drive: a guide for medical professionals into account when offering selfmonitoring of capillary blood glucose levels for adults with type 2 diabetes.

[2015, amended 2022]

1.6.2

Do not routinely offer self-monitoring of capillary blood glucose levels for adults with type 2 diabetes unless:
the person is on insulin or

there is evidence of hypoglycaemic episodes or

the person is on oral medication that may increase their risk of hypoglycaemia while driving or operating machinery or
they are pregnant or are planning to become pregnant (see

NICE's guideline on diabetes in pregnancy

[2015, amended 2022]

1.6.3

Consider short-term self-monitoring of capillary blood glucose levels in adults with type 2 diabetes, reviewing treatment as necessary:

when starting treatment with oral or intravenous corticosteroids or

to confirm suspected hypoglycaemia.

[2015, amended 2022]

1.6.4

Be aware that adults with type 2 diabetes who have acute intercurrent illness are at risk of worsening hyperglycaemia.

Review treatment as necessary.

[2015]

1.6.5

If adults with type 2 diabetes are self-monitoring their capillary blood glucose levels, carry out a structured assessment at least annually. The assessment should include:

the person's self-monitoring skills

the quality and frequency of testing

checking that the person knows how to interpret the blood glucose results and what action to take

the impact on the person's quality of life

the continued benefit to the person

the equipment used.

[2015, amended 2022]

1.7 Continuous glucose monitoring

Danh mục: medication

1.7.1

Offer intermittently scanned

continuous glucose monitoring

(isCGM, commonly referred to as 'flash') to adults with type 2 diabetes on

multiple daily insulin injections

if any of the following apply:

they have

recurrent hypoglycaemia

severe hypoglycaemia

they have impaired hypoglycaemia awareness

they have a condition or disability (including a learning disability or cognitive impairment) that means they cannot self-monitor their blood glucose by capillary blood glucose monitoring but could use an isCGM device (or have it scanned for them)

they would otherwise be advised to self-measure at least 8 times a day.

For guidance on continuous glucose monitoring (CGM) in pregnancy, see

NICE's guideline on diabetes in pregnancy

[2022]

1.7.2

Offer isCGM to adults with insulin-treated type 2 diabetes who would otherwise need help from a care worker or healthcare professional to monitor their blood glucose.

[2022]

1.7.3

Consider real-time continuous glucose monitoring (rtCGM) as an alternative to isCGM for adults with insulin-treated type 2 diabetes if it is available for the same or lower cost.

[2022]

1.7.4

CGM should be provided by a team with expertise in its use, as part of supporting people to self-manage their diabetes.

[2022]

1.7.5

Advise adults with type 2 diabetes who are using CGM that they will still need to take capillary blood glucose measurements (although they can do this less often). Explain that is because:

they will need to use capillary blood glucose measurements to check the accuracy of their CGM device

they will need capillary blood glucose monitoring as a back-up (for example when their blood glucose levels are changing quickly or if the device stops working).

Provide them with enough test strips to take capillary blood glucose measurements as needed.

[2022]

1.7.6

If a person is offered rtCGM or isCGM but cannot or does not want to use any of these devices, offer capillary blood glucose monitoring.

[2022]

1.7.7

Ensure CGM is part of the education provided to adults with type 2 diabetes who are using it (see the

Danh mục: medication

section on education

[2022]

1.7.8

Monitor and review the person's use of CGM as part of reviewing their diabetes care plan (see the section on individualised care

[2022]

1.7.9

If there are concerns about the way a person is using the CGM device:

ask if they are having problems using their device

look at ways to address any problems and concerns to improve their use of the device, including further education and emotional and psychological support.

[2022]

1.7.10

Commissioners, providers and healthcare professionals should address inequalities in CGM access and uptake by: monitoring who is using CGM

identifying groups who are eligible but who have a lower uptake

making plans to engage with these groups to encourage them to use CGM.

[2022]

Why the committee made these recommendations

The committee discussed how continuous glucose monitoring (CGM) could potentially be useful for many people with type 2 diabetes. They were aware of examples from current practice in which adults who have insulin-treated type 2 diabetes and use intermittently scanned CGM (isCGM) have had good outcomes. Because of the additional cost associated with CGM and the large number of adults with type 2 diabetes, the committee used both the evidence and their clinical experience to decide who would gain the most benefit from using CGM.

There was evidence that isCGM was cost effective for adults with type 2 diabetes using insulin, but no evidence for populations not using insulin, so the committee agreed to restrict their recommendations to adults using insulin.

People who have recurrent or severe hypoglycaemic events were identified as one of the groups likely to benefit most from isCGM, because hypoglycaemic events were considered to be one of the most important and concerning outcomes for adults with type 2 diabetes who are using insulin. The committee decided that the number of hypoglycaemic events was a more effective indicator of someone who would benefit from isCGM than specific HbA1c target values, because target values can vary between people. The evidence also indicated minimal effects of isCGM on HbA1c values.

The committee agreed that people with impaired hypoglycaemic awareness would also benefit from isCGM. However, they did not recommend specific methods for assessing impaired hypoglycaemic awareness. This is because validated methods for assessing impaired hypoglycaemic awareness in people with type 2 diabetes (such as the GOLD or Clarke scores) are not always available in primary care.

People who use insulin and have a condition or disability that restricts their ability to self-monitor blood glucose levels should also be offered isCGM. This is because having access to isCGM means they will no longer have to rely on others to monitor their diabetes, potentially increasing their independence.

People who need help from a care worker or other healthcare professional to administer their insulin injections should also be offered isCGM, even if they only use once-daily insulin injections. isCGM will help care workers to record a person's blood glucose levels quickly. And for people who have multiple home care visits per day, blood glucose levels can be recorded at each visit. This will help them to adjust their insulin levels to reduce the risk of hypoglycaemic events

Danh mục: medication

between home visits. It may also reduce the number of hospital admissions for this group.

The committee discussed how short-term use of isCGM may still be useful for some people. It may help people to understand when they have hypoglycaemic episodes, which would help them to develop a more effective treatment plan.

There was no evidence that rtCGM was cost effective for people with type 2 diabetes, so the committee agreed it could not be recommended for all adults with type 2 diabetes (whether or not they used insulin). They noted, however, that prices of rtCGM have reduced over the past few years, and if this continues to happen there may come a time when it is no more expensive than isCGM. At this point, rtCGM would be an appropriate alternative for people who meet the criteria for isCGM.

The committee did not make a recommendation on using specific devices because CGM technologies are changing very quickly and this recommendation would soon be out of date. Local healthcare services are better placed to assess which devices are evidence-based and suitable for use at any given time.

The committee discussed how self-monitoring of blood glucose should still take place, albeit less frequently, even when a person is using CGM. The ability to self-monitor blood glucose levels allows people to ensure the accuracy of the CGM device. The committee also recommended keeping capillary blood glucose monitoring as a back-up for situations such as when the technology fails.

The committee decided to highlight that CGM should be provided by a team who have expertise in its use. To ensure that CGM is effective, healthcare professionals need to have the skills to interpret and communicate the data effectively. As well as healthcare professionals having a clear understanding of CGM, it is also crucial that people with type 2 diabetes who are using CGM have education about the technology. This will increase the likelihood that people will scan and report the results frequently, allowing people to understand and manage their diabetes effectively.

Although many people will choose CGM if offered, there are some people who either cannot be offered it or do not want to use it. Because it is still important for these people to monitor their blood glucose levels, the committee made a recommendation to reinforce the importance of offering capillary blood glucose monitoring instead.

The committee did not make a recommendation on how long CGM should be used because there was no evidence on this, and they did not want to stop people accessing CGM for short periods if they and their healthcare professional thought they could benefit from this. Using CGM for a short period of time may help some people to understand when they have hypoglycaemic episodes, thereby helping them to develop a more effective treatment plan.

Despite the positive recommendations on CGM, the committee were concerned that existing health inequalities may still lead to lower uptake of CGM in some groups of people. To address this, the committee made a recommendation outlining actions for commissioners, providers and healthcare professionals.

The committee highlighted the importance of routinely reviewing a person's use of CGM. This will establish whether it is providing clinical benefits and whether the monitor is being used correctly. Making people aware that their use of CGM will be continually reviewed is important so they know it is not a guaranteed long-term option.

The committee also made a

recommendation for research on using routinely collected real-world data to assess the effectiveness and cost effectiveness of CGM

. They agreed that this has the potential to show the direct effects of the technology used by people with type 2 diabetes instead of interpreting it through the results of clinical trials. Increased monitoring of routine healthcare data including registries and audits would ensure that findings from a broader population are captured.

Full details of the evidence and the committee's discussion are in

evidence review C: continuous glucose monitoring in adults with type 2 diabetes

Danh mục: medication

How the recommendations might affect practice

The recommendations are likely to increase the number of adults with type 2 diabetes who are offered CGM, particularly those who have issues with hypoglycaemia. This will have associated cost implications:

It may save the NHS time, because healthcare professionals do not have to meet people who are using CGM as often as people who use capillary blood glucose monitoring.

There should be fewer hypoglycaemic events to manage.

The committee did not expect a significant resource impact related to education and monitoring for the CGM devices.

1.8 Hyperglycaemia

1.8.1

If an adult with type 2 diabetes has symptoms of hyperglycaemia, consider insulin (see the section on insulin-based treatments

) or a sulfonylurea, and review treatment when blood glucose is within the targets set for the person.

[2015]